

Lecture 10 – Completeness and Lehmann–Scheffé

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Outline:

10.1 Which sufficient statistic should we condition on?	1
10.2 When does conditioning give the UMVUE?	3
10.3 Completeness and minimal sufficiency	4
10.4 Establishing completeness	5
10.5 Complete sufficient statistics in exponential families	6
10.6 What the theorem does and does not guarantee	7

10.1 Which sufficient statistic should we condition on?

In Lecture 9, we learned how to improve an unbiased estimator U of $g(\theta)$: condition on a sufficient statistic T . The estimator $\mathbb{E}[U \mid T]$ is still unbiased and has variance no larger than U . But *which sufficient statistic should we choose?* For example, the full data $T(\underline{X}) = \underline{X}$ are sufficient, but

$$\mathbb{E}[U \mid \underline{X}] = U,$$

so this choice does not improve our estimator at all.

Coarser statistics give more averaging

Suppose $S = r(T)$ for some function r . We say that S is **coarser** than T : knowing T tells us S , but knowing S need not tell us T . Each group of datasets with the same value of S is a union of groups with the same value of T .

The estimator $\mathbb{E}[U \mid T]$ assigns an average value of U to each of the finer groups. Conditioning this estimator on S averages those group means together, using their conditional probabilities as weights. This removes variation between the finer groups within each coarser group. Averaging in these two stages gives the same result as averaging U directly within each S -group.

Thus, we expect conditioning on S to reduce variance at least as much as conditioning on T . If S and T are both sufficient, both averages are valid estimators, so we should prefer the coarser statistic. We now prove these claims.

Theorem 1 (Tower property for nested conditioning). Take any $U, T, r(T)$ such that the below expectations are finite. Then

$$\boxed{\mathbb{E}[\mathbb{E}[U \mid T] \mid r(T)] = \mathbb{E}[U \mid r(T)]}.$$

Proof (optional). We give the proof in the case that U, T are discrete. Write $m(t) = \mathbb{E}[U \mid T = t]$, so that $\mathbb{E}[U \mid T] = m(T)$. Then we need to prove that for each possible value $r(T) = s$, it holds

$$\mathbb{E}[m(T) \mid r(T) = s] = \mathbb{E}[U \mid r(T) = s].$$

Let us simplify the lefthand side to show it equals the righthand side. By the definition of conditional expectation, we have

$$\mathbb{E}[m(T) \mid r(T) = s] = \sum_t m(t) \mathbb{P}(T = t \mid r(T) = s) = \sum_t m(t) \frac{\mathbb{P}(T = t, r(T) = s)}{\mathbb{P}(r(T) = s)}$$

Now, if t is such that $r(t) \neq s$, we have $\mathbb{P}(T = t, r(T) = s) = 0$. If t is such that $r(t) = s$, then $\mathbb{P}(T = t, r(T) = s) = \mathbb{P}(T = t, r(T) = r(t)) = \mathbb{P}(T = t)$. Therefore, the sum above simplifies to

$$\begin{aligned}
 \mathbb{E}[m(T) \mid r(T) = s] &= \sum_{t:r(t)=s} m(t) \frac{\mathbb{P}(T = t)}{\mathbb{P}(r(T) = s)} \\
 &= \sum_{t:r(t)=s} \left(\sum_u u \mathbb{P}(U = u \mid T = t) \right) \frac{\mathbb{P}(T = t)}{\mathbb{P}(r(T) = s)} && \text{(definition of } m(t)) \\
 &= \frac{1}{\mathbb{P}(r(T) = s)} \sum_u u \sum_{t:r(t)=s} \mathbb{P}(U = u \mid T = t) \mathbb{P}(T = t) && \text{(rearrange sums)} \\
 &= \frac{1}{\mathbb{P}(r(T) = s)} \sum_u u \sum_{t:r(t)=s} \mathbb{P}(U = u, T = t) && \text{(conditional prob. def)} \\
 &= \frac{1}{\mathbb{P}(r(T) = s)} \sum_u u \mathbb{P}(U = u, r(T) = s) && \text{(sum over } t \text{ with } r(t) = s) \\
 &= \sum_u u \mathbb{P}(U = u \mid r(T) = s) = \mathbb{E}[U \mid r(T) = s]. && \text{(conditional prob. def)}
 \end{aligned}$$

□

Corollary 1 (Variance decreases under coarser conditioning). For any $U, T, r(T)$ such that the below expectations are finite, it holds

$$\boxed{\text{Var}(\mathbb{E}[U \mid r(T)]) \leq \text{Var}(\mathbb{E}[U \mid T])}.$$

Proof. Apply the law of total variance to $W = \mathbb{E}[U \mid T]$, conditioning on $r(T)$:

$$\begin{aligned}
 \text{Var}(\mathbb{E}[U \mid T]) &= \text{Var}(W) && \text{(definition of } W) \\
 &= \text{Var}(\mathbb{E}[W \mid r(T)]) + \mathbb{E}[\text{Var}(W \mid r(T))] && \text{(total variance)} \\
 &\geq \text{Var}(\mathbb{E}[W \mid r(T)]) && \text{(variance} \geq 0) \\
 &= \text{Var}(\mathbb{E}[\mathbb{E}[U \mid T] \mid r(T)]) && \text{(definition of } W) \\
 &= \text{Var}(\mathbb{E}[U \mid r(T)]). && \text{(conditional tower property)}
 \end{aligned}$$

□

Conditioning on a minimal sufficient statistic

Recall that S is **minimal sufficient** if it is sufficient and is a function of every other sufficient statistic T . Consequently, for a fixed unbiased estimator U ,

$$\text{Var}_\theta(\mathbb{E}[U \mid S]) \leq \text{Var}_\theta(\mathbb{E}[U \mid T]) \quad \text{for every sufficient } T \text{ and every } \theta.$$

So a minimal sufficient statistic gives the greatest variance reduction available by conditioning this U on a sufficient statistic. The reduction need not be strict: if the finer group means are already equal within each coarser group, there is no additional variation to remove.

Example 1 (The Poisson estimator from Lecture 9). Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\theta)$, with $n \geq 2$ and $\theta > 0$. We started with the proportion of zeros,

$$U = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{X_i=0\}}, \quad \mathbb{E}_\theta[U] = e^{-\theta},$$

and obtained

$$\mathbb{E}[U | T] = \left(1 - \frac{1}{n}\right)^T, \quad T = \sum_{i=1}^n X_i.$$

The statistic T is minimal sufficient. Thus no other choice of sufficient statistic can give a smaller variance by conditioning this same U . But have we found the UMVUE? We still need to compare with estimators obtained from *other* unbiased starting estimators.

10.2 When does conditioning give the UMVUE?

We have addressed which sufficient statistic to condition on. The remaining issue is that two different unbiased estimators U, V might give two different answers $\mathbb{E}[U | T]$ and $\mathbb{E}[V | T]$, even if T is minimal sufficient.

What additional property of T would make conditioning conclusive? It would be enough if there were **only one unbiased function of T for our target**. Then every unbiased estimator would condition down to the same answer.

When we say two estimators are equal, we mean that they give the same answer with probability 1 for every θ . We allow them to differ on an event of probability zero. We explain this convention with an example below. In the variance comparisons, we assume the estimators have finite variance.

Proposition 1. Suppose T is sufficient and $h(T)$ is unbiased for $g(\theta)$. If $h(T)$ is the *only* unbiased function of T for $g(\theta)$, then $\mathbb{E}[U | T] = \mathbb{E}[V | T] = h(T)$ for any two unbiased estimators U, V of $g(\theta)$, and $h(T)$ is the UMVUE.

Proof. Since T is sufficient, $\mathbb{E}[U | T]$ and $\mathbb{E}[V | T]$ are functions of T only, and by the tower property, they are both unbiased estimator of $g(\theta)$. Therefore by the condition of the proposition, they both are equal to one another and to $h(T)$. To prove $h(T)$ is the UMVUE, it suffices to show $\text{Var}_\theta(h(T)) \leq \text{Var}_\theta(U)$ for any unbiased estimator U of $g(\theta)$. But since conditioning reduces variance and $h(T) = \mathbb{E}[U | T]$, we know that

$$\text{Var}_\theta(h(T)) = \text{Var}_\theta(\mathbb{E}[U | T]) \leq \text{Var}_\theta(U),$$

as desired. □

Completeness

We now formalize the desired property of T above, that there is only one unbiased estimator of $g(\theta)$ that is a function of T . Suppose $h_1(T)$ and $h_2(T)$ are both unbiased for $g(\theta)$. Their difference satisfies

$$\mathbb{E}_\theta[h_1(T) - h_2(T)] = 0 \quad \text{for every } \theta.$$

We want this to force $h_1(T) - h_2(T) = 0$ with probability 1. “With probability 1” allows the formulas $h_1(t)$ and $h_2(t)$ to differ at individual points t , for which $\mathbb{P}(T = t) = 0$. For example, if T has a continuous distribution for every θ , then $\mathbb{P}_\theta(T = t_0) = 0$ for any fixed value t_0 . If $h_1(t) = h_2(t)$ everywhere except at t_0 , we still have $\mathbb{P}_\theta(h_1(T) = h_2(T)) = 1$: changing the answer at that single value does not change the estimator’s expectation or variance. For a discrete T , a discrepancy at a value with positive probability would matter. This motivates the following definition.

Definition 1 (Completeness). A statistic T is **complete** for the model if, for every function a whose expectation is finite for every θ ,

$$\mathbb{E}_\theta[a(T)] = 0 \text{ for every } \theta \implies a(T) = 0 \text{ with prob 1 for every } \theta.$$

In words, completeness says there are no nonzero functions of T whose expectation is zero for every value of θ . Without completeness, there can be a non-zero function $a(T) \neq 0$ whose expectation is zero for every θ , so adding $a(T)$ to an original unbiased estimator $h(T)$ of $g(\theta)$ would give a different unbiased estimator $a(T) + h(T)$.

Corollary 2 (Uniqueness of unbiased functions of a complete statistic). If T is complete and $h_1(T), h_2(T)$ are unbiased for the same $g(\theta)$, then $\mathbb{P}_\theta(h_1(T) = h_2(T)) = 1$ for every θ .

Proof. Apply completeness to $a(T) = h_1(T) - h_2(T)$. $\square$

Theorem 2 (Lehmann–Scheffé). Let T be sufficient and complete. If $h(T)$ is unbiased for $g(\theta)$ and has finite variance for every θ , then $h(T)$ is an UMVUE of $g(\theta)$. Furthermore, there is a single, essentially unique UMVUE, which is defined to mean that all UMVUEs differ at most on an event of probability zero.

Example 2 (Differing on event of probability zero). Suppose we have an estimator $U_1 = U_1(\underline{X})$ for a sample $\underline{X}$ drawn from a continuous distribution. Define a new estimator U_2 as

$$U_2(\underline{X}) = \begin{cases} U_1(\underline{X}) & \text{if } \underline{X} \neq \underline{0}, \\ U_1(\underline{0}) + 1, & \text{if } \underline{X} = \underline{0}. \end{cases}$$

Then

$$\mathbb{P}_\theta(U_1 \neq U_2) = \mathbb{P}_\theta(\underline{X} = \underline{0}) = 0 \quad \text{for every } \theta,$$

provided the distributions are indeed continuous for every value of θ . Thus $U_1 = U_2$ up to a set of probability zero, so we treat them as the same.

Proof. Completeness gives uniqueness among unbiased functions of T , by Corollary 2. Sufficiency and Proposition 1 then give that $h(T)$ has smallest variance among all unbiased estimators and is therefore an UMVUE. Furthermore, $\mathbb{E}[U | T] = h(T)$, so $\mathbb{E}[U | T]$ is an UMVUE as well.

To prove there cannot be multiple UMVUEs, suppose V is another UMVUE. We have $\mathbb{E}[V | T] = h(T)$, and both UMVUEs must have the same variance. Total variance therefore gives

$$0 = \text{Var}_\theta(V) - \text{Var}_\theta(h(T)) = \text{Var}_\theta(V) - \text{Var}_\theta(\mathbb{E}[V | T]) = \mathbb{E}_\theta[\text{Var}(V | T)].$$

Thus $\text{Var}(V | T) = 0$ with probability 1 for every θ , which implies $V = \mathbb{E}[V | T] = h(T)$ with probability 1 for every θ . Thus the UMVUE is essentially unique. $\square$

The most convenient way to use the theorem is to find *any* unbiased estimator and condition it on T . The result must be the UMVUE.

10.3 Completeness and minimal sufficiency

Minimal sufficiency says we cannot compress a sufficient statistic further, except by an equivalent representation, while keeping sufficiency. Completeness says there is at most one unbiased function of the statistic for each target. These are different claims: maximal sufficient compression need not give uniqueness of unbiased answers.

Example 3 (Minimal sufficient, but not complete). Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(\theta, \theta + 1)$, where $\theta \in \mathbb{R}$ and $n \geq 2$. We have shown before that $T = (X_{(1)}, X_{(n)})$ is minimal sufficient. But T is not complete. To show this, we use that $\text{Unif}(\theta, \theta + 1)$ is a location family. Write $X_i = \theta + Z_i$ with $Z_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, 1)$. We know that $h(T) = X_{(n)} - X_{(1)} = Z_{(n)} - Z_{(1)}$ is *ancillary*: its distribution does not depend on θ . Thus for every θ , we have $\mathbb{E}_\theta[X_{(n)} - X_{(1)}] = c$ for some constant c (working out the math gives $c = (n - 1)/(n + 1)$). Therefore, the nonzero function $a(T) = X_{(n)} - X_{(1)} - c$ has $\mathbb{E}_\theta[a(T)] = 0$ for all θ , so T is not complete.

Theorem 3 (Complete and sufficient implies minimal sufficient). Suppose T is a statistic or a vector of statistics. If T is sufficient and complete, then it is minimal sufficient, using our convention that equalities may ignore events of probability zero.

Proof when T has finite variance. We give the proof assuming that T (or each component, if T is a vector) has finite variance for every θ . Let S be any sufficient statistic. We must show that T is a function of S .

First suppose T is a univariate statistic. Define $g(\theta) = \mathbb{E}_\theta[T]$. Then T is an unbiased function of the complete sufficient statistic T , so Lehmann–Scheffé implies that T is the UMVUE of $g(\theta)$.

Since S is sufficient, Rao–Blackwell gives another unbiased estimator $\mathbb{E}[T | S]$ with

$$\text{Var}_\theta(\mathbb{E}[T | S]) \leq \text{Var}_\theta(T).$$

But T is the UMVUE, so equality holds, and total variance gives

$$0 = \text{Var}_\theta(T) - \text{Var}_\theta(\mathbb{E}[T | S]) = \mathbb{E}_\theta[\text{Var}_\theta(T | S)].$$

It follows that $\text{Var}_\theta(T | S) = \mathbb{E}_\theta[(T - \mathbb{E}[T | S])^2 | S] = 0$ with probability 1 for every θ , which implies $T = \mathbb{E}[T | S]$ with probability 1 for every θ . Sufficiency of S ensures that $\mathbb{E}[T | S]$ does not depend on θ . Thus T is a function of S , as required.

For a vector $T = (T_1, \dots, T_k)$, each T_j is an unbiased function of the complete sufficient statistic T for the target $g_j(\theta) = \mathbb{E}_\theta[T_j]$. Apply the same argument to each coordinate to obtain $T_j = \mathbb{E}[T_j | S]$. Thus the whole vector is a function of S . $\square$

We therefore have

$$\boxed{\text{complete and sufficient} \implies \text{minimal sufficient}},$$

but the converse fails, as Example 3 shows. A complete sufficient statistic gives both the coarsest sufficient representation and uniqueness of unbiased functions.

Proposition 2 (Functions preserve completeness). If T is complete and $S = r(T)$, then S is complete.

Proof. If $\mathbb{E}_\theta[a(S)] = 0$ for every θ , then $\mathbb{E}_\theta[a(r(T))]$ is 0 for every θ . Apply completeness of T to the function $t \mapsto a(r(t))$ to obtain $a(S) = a(r(T)) = 0$ with probability 1 for every θ . $\square$

Coarsening preserves completeness, but it need not preserve sufficiency. For example, a constant statistic is complete, but generally not sufficient. A one-to-one transformation preserves both properties, since it retains exactly the same information.

10.4 Establishing completeness

We show how to do this in an example.

Example 4 (Uniform scale model). Suppose $X_i \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$, $\theta > 0$, and let $T = X_{(n)}$. We have shown that T is sufficient, in fact, it is minimal sufficient. First, one can show (do this as an exercise) that the density of T is the following:

$$f_T(t) = \frac{n}{\theta^n} t^{n-1} \mathbb{1}_{\{0 < t < \theta\}}.$$

Let a be any function whose expectation is finite for every θ , and suppose $\mathbb{E}_\theta[a(T)] = 0$ for every $\theta > 0$. Then

$$0 = \frac{n}{\theta^n} \int_0^\theta a(t) t^{n-1} dt \implies \int_0^\theta a(t) t^{n-1} dt = 0 \quad \text{for every } \theta > 0.$$

The calculus idea is to differentiate this identically zero integral with respect to its upper limit:

$$0 = \frac{d}{d\theta} \int_0^\theta a(t)t^{n-1} dt = a(\theta)\theta^{n-1}.$$

This holds by the usual fundamental theorem of calculus. We conclude from here that $a(\theta) = 0$ for all $\theta > 0$. Thus $a(T) = 0$ with probability 1, since $\mathbb{P}_\theta(T = 0) = 0$. This proves completeness.

Let us now use completeness to find the UMVUE of θ . First, note that

$$\mathbb{E}_\theta[T] = \frac{n}{\theta^n} \int_0^\theta t^n dt = \frac{n}{n+1}\theta.$$

Therefore, $\frac{n+1}{n}T$ is an unbiased estimator of θ , so by the Lehmann-Scheffe theorem, $\mathbb{E}[\frac{n+1}{n}T \mid T] = \frac{n+1}{n}T$ is itself the UMVUE of θ .

$$\frac{n+1}{n}X_{(n)} \text{ is the UMVUE of } \theta.$$